Tertiary Infotech Academy Pte. Ltd.

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LESSON PLAN

For

Blender Masterclass

Course Code: C140

Conducted by

Tertiary Infotech Academy Pte. Ltd.

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Version 1.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	30 September 2026	Initial release of the Blender Masterclass courseware: four topics, eight connected Blender labs.	Tertiary Infotech Academy

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Course Overview

A two-day, 15-hour course in which beginners interpret a creative brief, build and animate a Blender prototype, then optimise, render and export a verified 3D asset.

Learning Outcomes

- LO1: Analyse a creative brief and establish a suitable polygon budget.
- LO2: Develop 3D models and prototypes in Blender.
- LO3: Animate 3D models using suitable rigging, keyframe and presentation methods.
- LO4: Optimise, light, render and export 3D models for the intended use.

Daily Schedule

Each day runs 9:30am – 5:30pm: 7.5 instructional hours with a single 30-minute lunch.

Day	Time	Duration	Topic / Lab	Method	Slides
1	09:30–10:30	1h	Course administration and Topic 1 introduction	Briefing, demonstration and discussion	Slides 2–14
1	10:30–12:15	1h 45m	Topic 1: Get Started with Blender and the Creative Brief	Concept workshop and Lab 1	Slides 15–34
1	12:15–12:45	0h 30m	Lunch break	Break	—
1	12:45–14:45	2h	Topic 1 application and Topic 2 introduction	Lab 2 and guided modelling	Slides 35–41
1	14:45–17:30	2h 45m	Topic 2: Develop 3D Models and Prototypes	Demonstration, Labs 3–4 and review	Slides 42–67
2	09:30–11:30	2h	Topic 3: Animate and Present the 3D Prototype	Concept demonstration and coached practice	Slides 68–85
2	11:30–12:30	1h	Topic 3 application	Lab 5 and peer review	Slides 86–89
2	12:30–13:00	0h 30m	Lunch break	Break	—
2	13:00–14:45	1h 45m	Topic 3 application (continued)	Lab 6 and peer review	Slides 90–94
2	14:45–16:45	2h	Topic 4: Optimise, Render and Export the Model	Concept demonstration and Lab 7	Slides 95–116
2	16:45–17:30	0h 45m	Capstone release and course wrap-up	Lab 8 completion, recap and course feedback	Slides 117–127

Topic-by-Topic Breakdown

Topic 1: Get Started with Blender and the Creative Brief · Slides 13–39

Analyse requirements · establish budgets · control the scene

Alignment: LO1

Lab 1: Navigate Blender and organise a production scene · Slides 31–34

Navigate the interface, distinguish Object and Edit modes, organise objects and preserve a clean baseline.

Lab 2: Interpret a fish brief, budget polygons and build a blockout · Slides 35–38

Translate the brief into criteria, allocate a polygon budget and create a modifier-led blockout.

Topic 2: Develop 3D Models and Prototypes · Slides 40–67

Block out form · sculpt detail · refine topology

Alignment: LO2

Lab 3: Model and sculpt the production goldfish · Slides 59–62

Develop primary and secondary forms with suitable Edit Mode and sculpting tools.

Lab 4: Repair topology, normals and deformation zones · Slides 63–66

Diagnose and repair the supplied mesh, then prove the revised prototype meets its criteria.

Topic 3: Animate and Present the 3D Prototype · Slides 68–94

Rig movement · keyframe action · surface and present

Alignment: LO3

Lab 5: UV unwrap and surface the goldfish · Slides 86–89

Create controlled UVs, textures and a Principled BSDF material that remains readable under neutral light.

Lab 6: Rig and animate a looping swim cycle · Slides 90–93

Build a simple armature, validate deformation and animate a clean loop using controlled curves.

Topic 4: Optimise, Render and Export the Model · Slides 95–121

Frame the asset · validate quality · deliver reliably

Alignment: LO4

Lab 7: Create bubbles, light the scene and present for review · Slides 113–116

Add a lightweight bubble/force effect, build purposeful lighting and present a review decision.

Lab 8: Optimise, render and export the fish-bowl capstone · Slides 117–120

Optimise the full scene, render final evidence, export the intended asset and validate the package by round trip.

Resources Required

- Blender 5.2 LTS installed on Windows or macOS laptop
- Course lab folders and supplied Blender sample assets
- Presentation software for lab evidence
- Internet access for official Blender references and course materials